

Markov Chains (AI HL)

ANSWER KEY



Constructing a transition matrix

- 1 A weather model has two states, Sunny (S) and Rainy (R). From Sunny: 80% stay Sunny, 20% become Rainy. From Rainy: 40% become Sunny, 60% stay Rainy. Write down the transition matrix T , using columns for the 'from' state in the order (S, R).
 T has rows (0.8, 0.4) and (0.2, 0.6) — column 1 (from Sunny) is $(0.8, 0.2)^T$ and column 2 (from Rainy) is $(0.4, 0.6)^T$, each summing to 1.

Applying the transition matrix once

- 2 Today is sunny, so $\mathbf{p}_0 = (1, 0)^T$ (order S, R). Using the matrix T from the previous question, find $\mathbf{p}_1 = T\mathbf{p}_0$, the probability distribution for tomorrow.
 $\mathbf{p}_1$ is simply column 1 of T : $\mathbf{p}_1 = (0.8, 0.2)^T$. So $P(\text{sunny tomorrow}) = 0.8$ and $P(\text{rain tomorrow}) = 0.2$.

Applying the transition matrix twice

- 3 Using $\mathbf{p}_1 = (0.8, 0.2)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
Row 1: $0.8(0.8) + 0.4(0.2) = 0.64 + 0.08 = 0.72$. Row 2: $0.2(0.8) + 0.6(0.2) = 0.16 + 0.12 = 0.28$.
So $\mathbf{p}_2 = (0.72, 0.28)^T$.

Finding the steady-state vector

- 4 Find the steady-state vector $\mathbf{s} = (s_1, s_2)^T$ for the transition matrix T above, satisfying $T\mathbf{s} = \mathbf{s}$ and $s_1 + s_2 = 1$.
 $T\mathbf{s} = \mathbf{s}$ gives $0.8s_1 + 0.4s_2 = s_1$, i.e. $0.4s_2 = 0.2s_1$. So $s_2 = 0.5s_1$. Substituting into $s_1 + s_2 = 1$ gives $s_1 + 0.5s_1 = 1$. So $s_1 = 2/3$, and $s_2 = 1/3$.

Explaining why a Markov chain is regular

- 5 Explain why the Markov chain above converges to a unique steady state regardless of the initial distribution.
Every entry of the transition matrix T is strictly positive (no transition probability is zero), so the chain is regular. A regular chain's distribution $\mathbf{p}_n$ converges to the same steady-state vector $\mathbf{s}$ as $n \rightarrow \infty$, whatever the starting distribution $\mathbf{p}_0$ is.

Computing and interpreting T^2

- 6 Using the transition matrix T with rows $(0.8, 0.4)$ and $(0.2, 0.6)$, find T^2 and interpret its $(1,1)$ entry. T^2 has entries: $(1, 1) = 0.8(0.8) + 0.4(0.2) = 0.72$; $(1, 2) = 0.8(0.4) + 0.4(0.6) = 0.56$; $(2, 1) = 0.2(0.8) + 0.6(0.2) = 0.28$; $(2, 2) = 0.2(0.4) + 0.6(0.6) = 0.44$. The $(1, 1)$ entry, 0.72, is the probability of being Sunny two days from now given it is Sunny today — matching $\mathbf{p}_2$'s first entry found earlier.
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Constructing a 3-state transition matrix

- 7 A game has three states: Win (W), Draw (D), Loss (L). From W: 50% stay W, 30% go to D, 20% go to L. From D: 40% go to W, 20% stay D, 40% go to L. From L: 10% go to W, 30% go to D, 60% stay L. Write down the transition matrix T , using columns for the 'from' state in the order (W, D, L). T has rows (in order W, D, L): row W = $(0.5, 0.4, 0.1)$; row D = $(0.3, 0.2, 0.3)$; row L = $(0.2, 0.4, 0.6)$. Each column sums to 1, e.g. column W: $0.5 + 0.3 + 0.2 = 1$.
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Applying the transition matrix to a 3-state chain

- 8 Today's state is Draw, so $\mathbf{p}_0 = (0, 1, 0)^T$ (order W, D, L). Using the matrix T above, find $\mathbf{p}_1 = T\mathbf{p}_0$. $\mathbf{p}_1$ is column D of T : $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$. So $P(W) = 0.4$, $P(D) = 0.2$, $P(L) = 0.4$.
- 9 Using $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
Row W: $0.5(0.4) + 0.4(0.2) + 0.1(0.4) = 0.2 + 0.08 + 0.04 = 0.32$. Row D:
 $0.3(0.4) + 0.2(0.2) + 0.3(0.4) = 0.12 + 0.04 + 0.12 = 0.28$. Row L:
 $0.2(0.4) + 0.4(0.2) + 0.6(0.4) = 0.08 + 0.08 + 0.24 = 0.40$. So $\mathbf{p}_2 = (0.32, 0.28, 0.40)^T$ (entries sum to 1.00, as expected).
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Defining the steady-state equation

- 10 State the equation that defines a steady-state vector $\mathbf{s}$ of a transition matrix T , and explain what it means for $\mathbf{s}$ to be an eigenvector of T with eigenvalue 1.
A steady-state vector satisfies $T\mathbf{s} = \mathbf{s}$, with the entries of $\mathbf{s}$ summing to 1. Saying $\mathbf{s}$ is an eigenvector of T with eigenvalue 1 means exactly this: applying the transition matrix to $\mathbf{s}$ leaves it completely unchanged, so once the distribution reaches $\mathbf{s}$ it stays there forever.
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Verifying a steady-state vector directly

- 11 Verify that $\mathbf{s} = (2/3, 1/3)^T$ is the steady-state vector for the weather transition matrix T above, by checking $T\mathbf{s} = \mathbf{s}$ directly.
- Row 1: $0.8(2/3) + 0.4(1/3) = 1.6/3 + 0.4/3 = 2/3$, matching s_1 . Row 2:
 $0.2(2/3) + 0.6(1/3) = 0.4/3 + 0.6/3 = 1/3$, matching s_2 . Since $T\mathbf{s} = \mathbf{s}$ and $2/3 + 1/3 = 1$, $\mathbf{s}$ is confirmed as the steady-state vector.
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The memoryless (Markov) property

- 12 Explain what is meant by saying a Markov chain has the 'memoryless' property.
- The probability of moving to any state at the next step depends only on the current state, not on the sequence of states that preceded it — the history of how the system arrived at its current state has no further effect on predicting its future.
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Constructing a new transition matrix

- 13 A machine is either Working (W) or Broken (B) each day. If Working today, there is a 90% chance it is Working tomorrow. If Broken today, there is a 30% chance it is repaired and Working tomorrow. Write down the transition matrix T , using columns for the 'from' state in the order (W, B).
- T has rows (in order W, B): row W = (0.9, 0.3); row B = (0.1, 0.7). Each column sums to 1: column W is $0.9 + 0.1 = 1$, and column B is $0.3 + 0.7 = 1$.
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Applying the new transition matrix once

- 14 The machine is Working today, so $\mathbf{p}_0 = (1, 0)^T$ (order W, B). Using the matrix T above, find $\mathbf{p}_1 = T\mathbf{p}_0$. $\mathbf{p}_1$ is column W of T : $\mathbf{p}_1 = (0.9, 0.1)^T$. So $P(\text{Working tomorrow}) = 0.9$.
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Applying the new transition matrix twice

- 15 Using $\mathbf{p}_1 = (0.9, 0.1)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
- Row W: $0.9(0.9) + 0.3(0.1) = 0.81 + 0.03 = 0.84$. Row B:
 $0.1(0.9) + 0.7(0.1) = 0.09 + 0.07 = 0.16$. So $\mathbf{p}_2 = (0.84, 0.16)^T$.
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Finding the steady state of the new chain

- 16 Find the steady-state vector $\mathbf{s} = (s_1, s_2)^T$ for the machine's transition matrix T , satisfying $T\mathbf{s} = \mathbf{s}$ and $s_1 + s_2 = 1$.
- $T\mathbf{s} = \mathbf{s}$ gives $0.9s_1 + 0.3s_2 = s_1$, i.e. $0.3s_2 = 0.1s_1$. So $s_2 = (1/3)s_1$. Substituting into $s_1 + s_2 = 1$ gives $s_1 + (1/3)s_1 = 1$, so $(4/3)s_1 = 1$, giving $s_1 = 3/4$ and $s_2 = 1/4$.
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Recognizing an invalid transition matrix

- 17** Explain why a matrix with rows $(0.9, 0.3)$ and $(0.05, 0.6)$ (order W, B) could NOT be a valid transition matrix.

Every column of a transition matrix must sum to exactly 1, since a system currently in a given state must move to some state next. Here column W sums to $0.9 + 0.05 = 0.95 \neq 1$, so 5% of the probability of leaving state W is unaccounted for — the matrix is invalid.

Absorbing states

- 18** A game has two states: Playing (P) and Game Over (G). From P, there is an 80% chance of remaining in P and a 20% chance of moving to G. From G, there is a 100% chance of remaining in G. Explain why G is called an 'absorbing state', and describe the long-run behaviour of the game.

G is absorbing because, once the system enters G, the transition probability of leaving G is 0 — it can never return to P. Since every step in P carries a 20% chance of moving into G and none of leaving it, the probability of still being in P shrinks every step; in the long run, the system is absorbed into G with probability approaching 1, and the probability of being in P approaches 0.

Interpreting the steady state as a long-run prediction

- 19** Using the steady-state vector $\mathbf{s} = (3/4, 1/4)^T$ found earlier for the machine example, predict the long-run fraction of days the machine is Working.

Since every entry of T is strictly positive, the chain is regular, so $\mathbf{p}_n \rightarrow \mathbf{s}$ as $n \rightarrow \infty$ regardless of the starting distribution. In the long run, the machine is Working on 75% of days.

Using a steady state to predict a count

- 20** A regular Markov chain's steady-state vector is $\mathbf{s} = (0.4, 0.35, 0.25)$ for three states A, B, C. Over 1000 days long after the chain has settled into steady state, roughly how many days would you expect to find the system in state B?

The expected number of days in state B is $0.35 \times 1000 = 350$ days.

Computing and interpreting T^2 for the new chain

- 21** Using the machine's transition matrix T with rows $(0.9, 0.3)$ and $(0.1, 0.7)$, find T^2 and interpret its $(2, 1)$ entry.
- T^2 entries: $(1, 1) = 0.9(0.9) + 0.3(0.1) = 0.84$; $(1, 2) = 0.9(0.3) + 0.3(0.7) = 0.48$;
 $(2, 1) = 0.1(0.9) + 0.7(0.1) = 0.16$; $(2, 2) = 0.1(0.3) + 0.7(0.7) = 0.52$ (columns check:
 $0.84 + 0.16 = 1$ and $0.48 + 0.52 = 1$). The $(2, 1)$ entry, 0.16, is the probability of being Broken two days from now, given the machine is Working today — matching $\mathbf{p}_2$'s second entry found earlier.
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Verifying the new chain's steady state directly

- 22** Verify that $\mathbf{s} = (3/4, 1/4)^T$ is the steady-state vector for the machine's transition matrix T (rows $(0.9, 0.3)$ and $(0.1, 0.7)$), by checking $T\mathbf{s} = \mathbf{s}$ directly.
- Row 1: $0.9(3/4) + 0.3(1/4) = 2.7/4 + 0.3/4 = 3/4$, matching s_1 . Row 2:
 $0.1(3/4) + 0.7(1/4) = 0.3/4 + 0.7/4 = 1/4$, matching s_2 . Since $T\mathbf{s} = \mathbf{s}$ and $3/4 + 1/4 = 1$, $\mathbf{s}$ is confirmed as the steady-state vector.